

25W1 3: Counterexamples for Extragradient and the Performance of Halpern Iteration on VI Benchmarks

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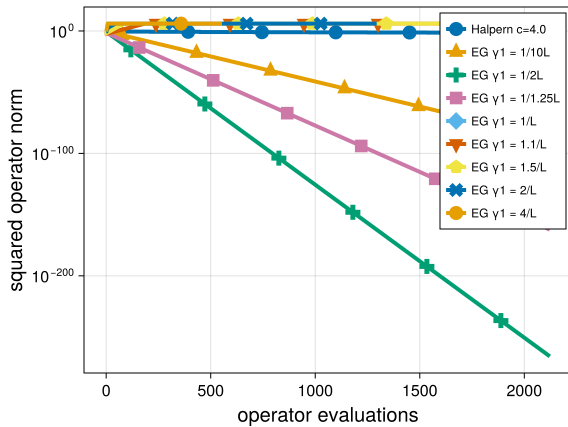
1 Introduction

To verify the counterexample on the extragradient (EG) method established in Theorem 4.3 of Gorbunov et al. [1], we first reproduce its divergence behavior on simple linear benchmark problems. In particular, we consider the operator $F(x) = Lx$, which diverges when $\gamma_1 > 1/L$, and the rotated operator $F(x) = LAx$ with A a 120° rotation matrix, which diverges when $0 < \gamma_1 < 1/L$. For all experiments, we set $\gamma_1 = \gamma_2$ to match the standard form of the method. In later sections, we also provide a comparison with the *stochastic Halpern* method.

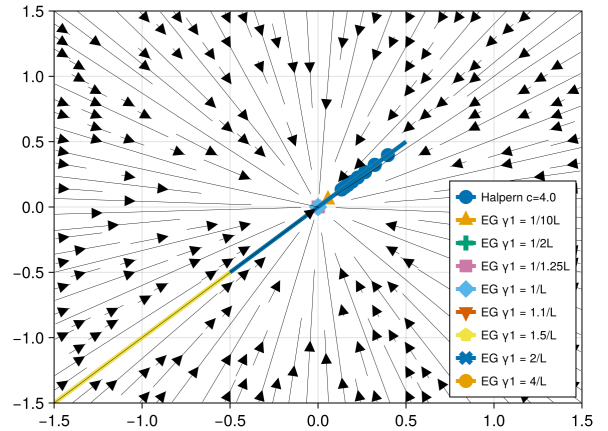
2 Experiments

2.1 Case I: Linear Operator $F(x) = Lx$

We first consider the simple linear operator $F(x) = Lx$ with $L = 1.0$, which corresponds to a one-dimensional rescaling. This setting directly illustrates the threshold behavior established in Theorem 4.3. The extragradient method is applied with $\gamma_1 = \gamma_2 = \eta$, and we vary η around the critical point $1/L = 1.0$.



(a) Squared operator norm vs. operator evaluations



(b) Iterate trajectories over the vector field

Figure 1: Extragradient on $F(x) = Lx$ with $\eta = 0.9/L$. Stable only for very small η , consistent with the threshold $\eta < 1/L$.

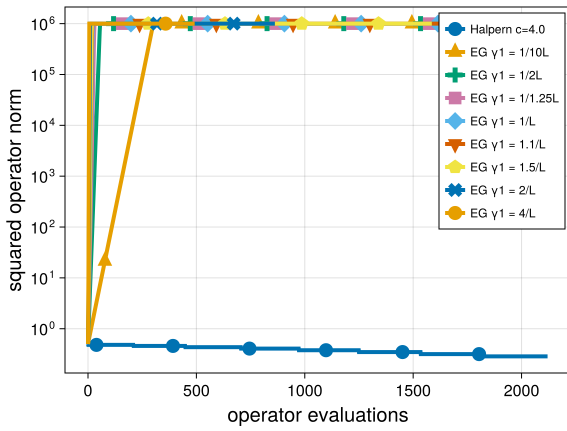
As predicted by theory, the extragradient method remains stable only when $\gamma_1 = \gamma_2 < 1/L$. Once the stepsize exceeds this critical value, the iterates diverge rapidly. In contrast, the Halpern method, although significantly slower, exhibits convergence in this regime.

2.2 Case II: Rotated Linear Operator $F(x) = LAx$

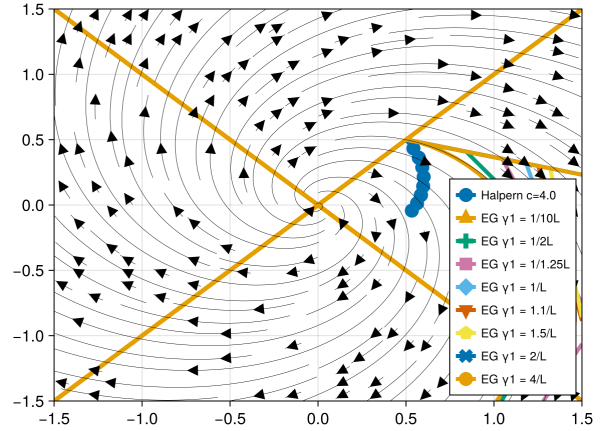
We next consider the rotated operator $F(x) = LAx$ with $L = 1.0$ and

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad \theta = \frac{2\pi}{3}.$$

This benchmark is the canonical counterexample from Theorem 4.3 of Gorbunov et al. [1], illustrating divergence behavior even when $\gamma_1 = \gamma_2$ is chosen below the $1/L$ threshold.



(a) Squared operator norm vs. operator evaluations



(b) Iterate trajectories over the vector field

Figure 2: Extragradient on $F(x) = LAx$ with $\eta = 0.9/L$. Unlike the Lx case, instability arises both when $0 < \eta < 1/L$ and when $\eta > 1/L$. Halpern, while much slower, still converges.

As predicted, the extragradient method fails to converge in this setting: divergence occurs not only for $\eta > 1/L$ but also in the regime $0 < \eta < 1/L$. This matches the theoretical lower bound $\rho \geq 1/(2L)$. By contrast, the Halpern method remains convergent, though significantly slower.

We then compare the deterministic and stochastic Halpern methods on the same problem.

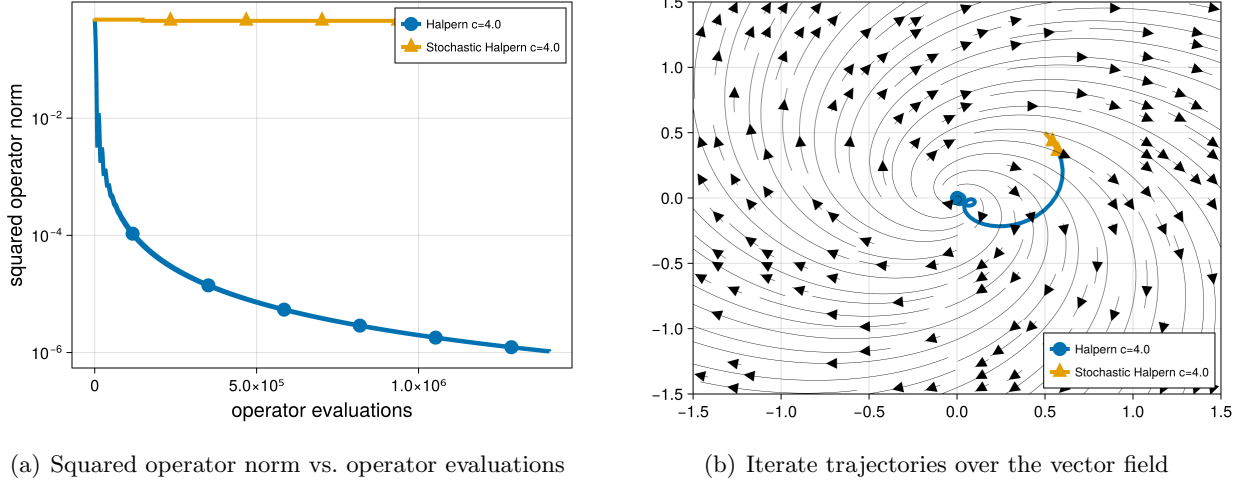


Figure 3: Halpern vs. stochastic Halpern on $F(x) = Lx$. Both methods converge, but the stochastic variant requires dramatically larger inner iterations.

Both Halpern and stochastic Halpern converge, but the stochastic variant is prohibitively slow. For example, the initial N_k was 104 for Halpern versus 74,054 for stochastic Halpern, and the second N_k grew to 627,855. With such enormous inner-loop sizes, even a budget of 1,403,820 gradient evaluations allowed only two outer iterations to be completed. This demonstrates the extreme inefficiency of the stochastic scheme in practice, despite its theoretical guarantees.

3 Conclusion

We verified Theorem 4.3 of Gorbunov et al. [1] by reproducing the divergence of the extragradient method on linear benchmarks. For $F(x) = Lx$, divergence occurs once $\eta > 1/L$, while for $F(x) = LAx$ it appears both below and above $1/L$. In contrast, the Halpern method always converges, though slowly. The stochastic Halpern method also converges but with impractically large inner iterations. Overall, our results confirm the theory and highlight the limitations of extragradient compared to the robustness of Halpern schemes.

References

- [1] Eduard Gorbunov, Adrien Taylor, Samuel Horváth, and Gauthier Gidel. Convergence of proximal point and extragradient-based methods beyond monotonicity: the case of negative comonotonicity, 2022.